

Supplementary Materials for

Decoding the multicellular ecosystem of lung adenocarcinoma manifested as pulmonary subsolid nodules by single-cell RNA sequencing

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Supplementary Figures

Figure S1

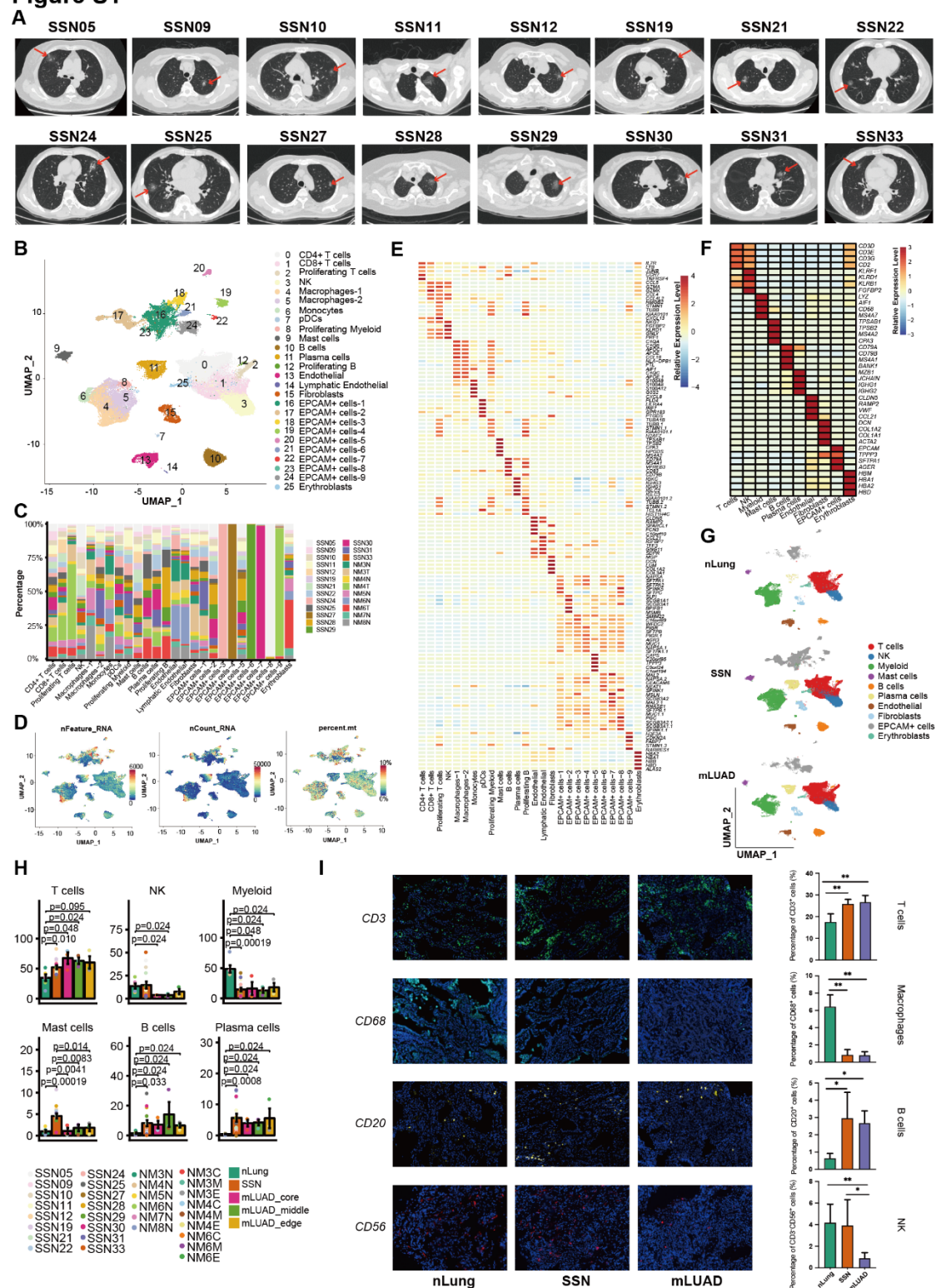
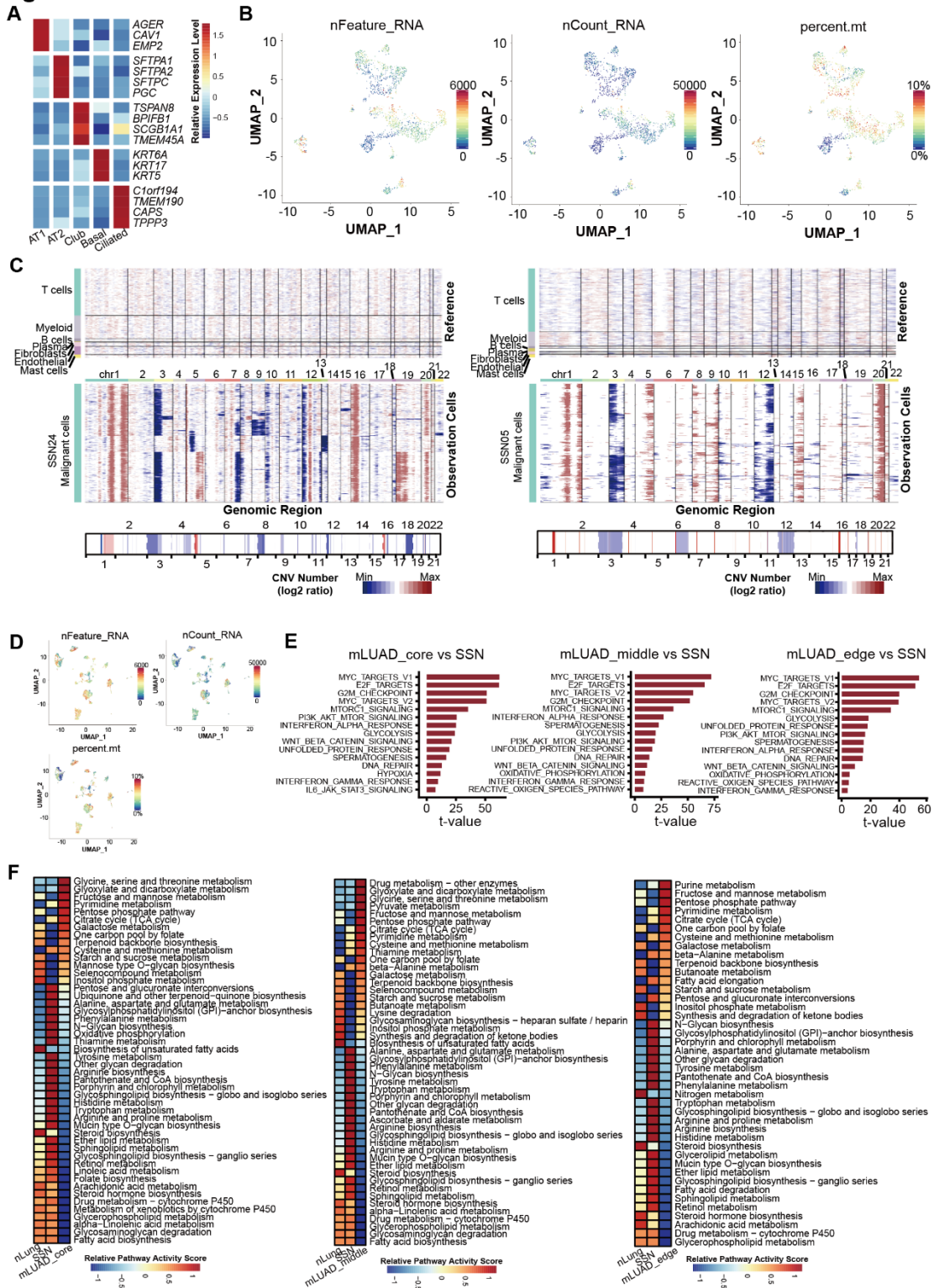


Figure S1. Radiological features of SSN samples and overview of clustering results, Related to Figure 1.

A. Radiological features of 16 SSN samples enrolled in the study. Red arrows indicating the tumor lesion. **B.** Clustering of 118,293 single cells from nLung (n=6), SSN (n=16) and

mLUAD (n=9) samples. Each dot corresponds to a single cell, colored according to cell subtypes. **C.** Sample distribution in each cell subtype. Each bar corresponds to one cell subtype, colored according to patients. **D.** UMAP plots of cells colored by number of genes detected, UMIs RNA, and percentage of mitochondrial genes. **E.** Heatmap of marker genes related to **(B)**, with the cell lineage of exemplary genes labelled (right). Columns denote cell subtypes and rows denote genes. **F.** Heatmap of marker genes related to 10 main cell lineages in **Figure 1B**. **G.** UMAP plots referring to **Figure 1D** showing the cellular compositions of nLung, SSN, and mLUAD. **H.** Percentages bar-plot similar to **Figure 1E**, but split mLUAD into different spatial positions. **I.** Representative images of the multi-plex staining and statistical analyses of cell percentages between different types of tissues. DAPI: blue, CD3: green, CD68: cyan, CD20: yellow; CD56: red; All differences with $P < 0.05$ are indicated; $*P < 0.05$; $**P < 0.01$; $***P < 0.001$; two-sided unpaired Wilcoxon rank-sum test was used for analysis.

Figure S2



(middle). CNVs of the sample were validated by whole exome sequencing analysis (bottom). The color shows the log2 CNV ratio. Red: amplifications; Blue: deletions. **D.** UMAP plot of malignant cells colored by number of genes detected, UMIs RNA and percentage of mitochondrial genes referring to **Figure 2E**. **E.** Similar to **Figure 2F**, but comparing SSN to mLUAD with different spatial positions. **F.** Similar to **Figure 2G**, but split mLUAD into different spatial positions.

Figure S3

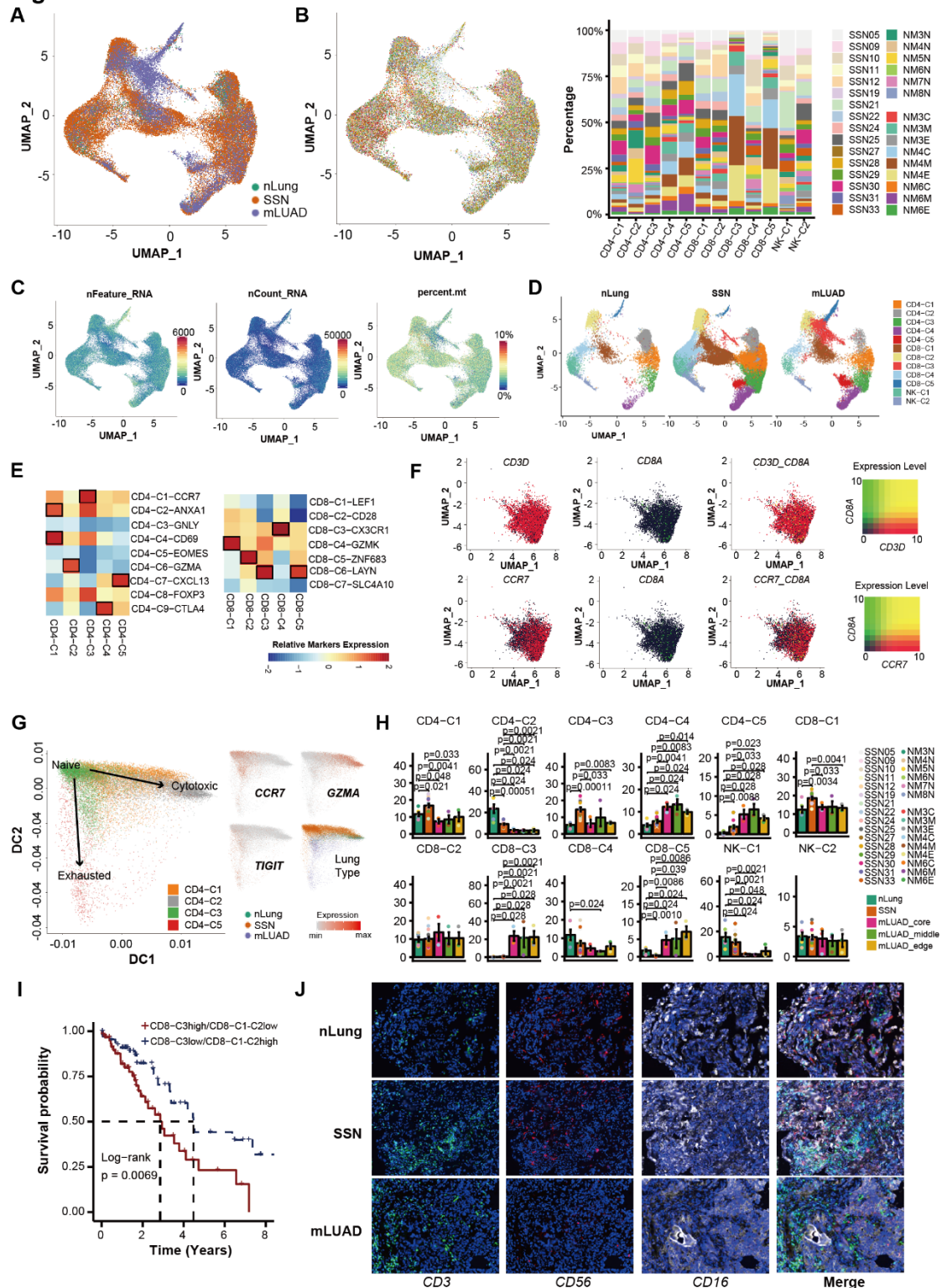


Figure S3. T and NK cells in nLung, SSN, and mLUAD, Related to Figure 3.

A. Distribution of different groups as shown in the UMAP plot. Each dot corresponds to a single cell, colored according to nLung, SSN and mLUAD. **B.** Sample distribution in the UMAP plot (left). Each bar corresponds to one cell subtype, colored according to sample name (right). **C.** UMAP plot of cells colored by number of genes detected, UMIs RNA and percentage of mitochondrial genes. **D.** Annotation of UMAP plots of T/NK subtypes split by

different groups. **E.** Heatmap showing the cluster similarity with Guo *et al.* by cluster marker expression levels. **F.** UMAP shows the co-expression of genes in CD4-C3 cluster. **G.** Development trajectory of CD4⁺ T cells inferred by Diffusion Map, colored by CD4⁺ T cell subtypes and expression of example genes. **H.** Percentages bar-plot similar to **Figure 3G**, but split mLUAD into different spatial positions. **I.** Kaplan-Meier plot showing that TCGA dataset lung adenocarcinoma patients with CD8-C3-high vs. CD8-C1/C2-low have shorter overall survival. The high and low groups are divided by the 60% quantile value of the mean expression of the above gene sets. P-value from univariate Cox regression. **J.** Representative multi-plex staining images of NK cells (CD3⁻CD56⁺) and NK subtype (NK-C1; CD3⁻CD56⁺CD16⁺) in nLung, SSN, and mLUAD. CD3: green, CD56: red, CD16: white.

Figure S4

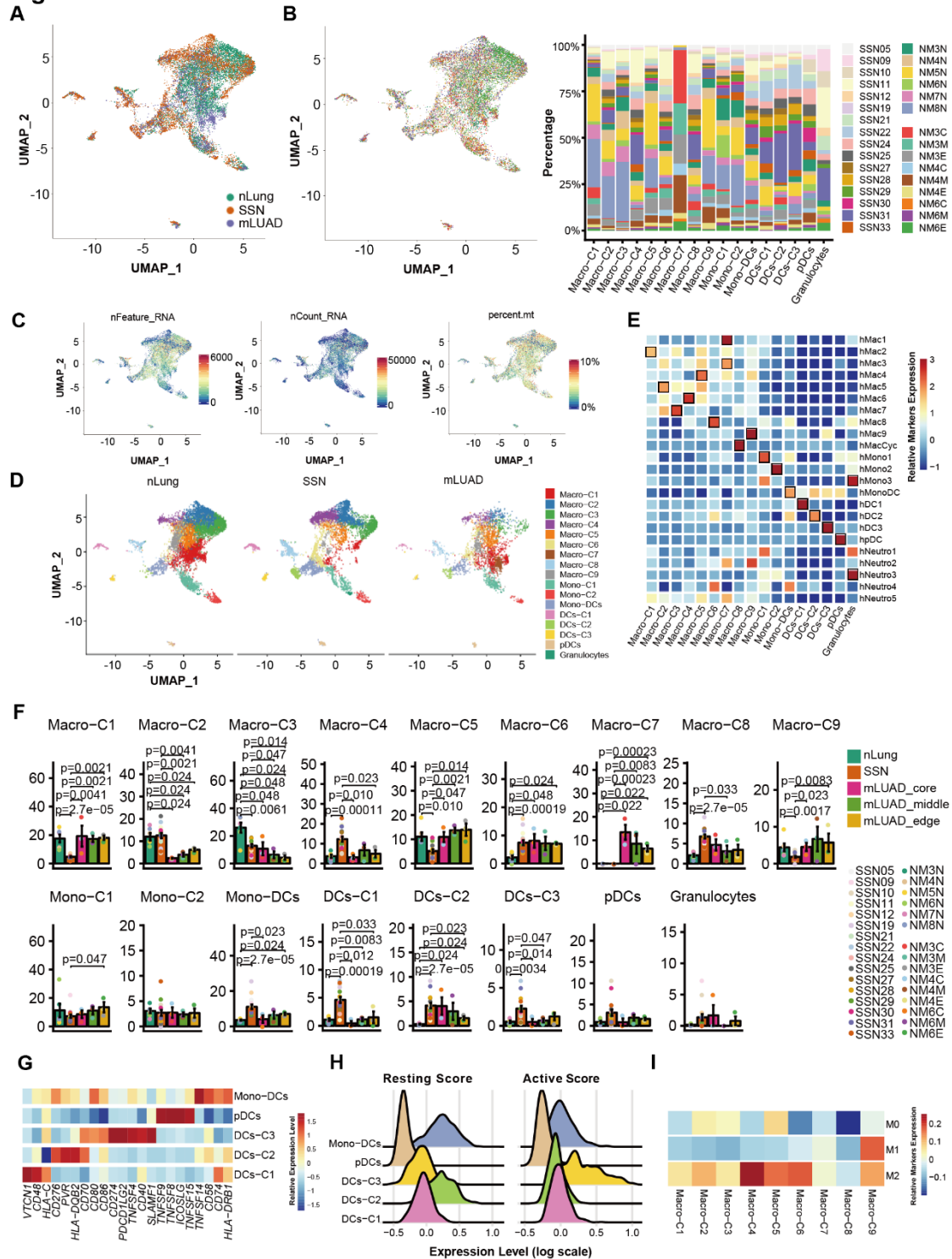


Figure S4. Myeloid cell subtypes in nLung, SSN, and mLUAD, Related to Figure 4.

A. Distribution of different groups as shown in the UMAP plot. Each dot corresponds to a single cell, colored according to nLung, SSN and mLUAD, respectively. **B.** Sample distribution in the UMAP plot (left). Each bar corresponds to one cell type cluster, colored according to sample name (right). **C.** UMAP plot of cells colored by number of genes detected, UMIs RNA and percentage of mitochondrial genes. **D.** Annotation of the UMAP plots of myeloid cell subtypes split by different groups. **E.** Heatmap showing cluster

similarity with Zilionis *et al.* by cluster marker expression levels. **F.** Percentages bar-plot similar to **Figure 4F**, but split mLUAD into different spatial positions. **G.** Heatmap of antigen presenting genes and their co-activators expression in DCs clusters. **H.** Ridge plots showing the distribution of resting and active state DC subtypes. **I.** Heatmap showing the relative expression levels of M0, M1 and M2 markers for different macrophage subtypes.

Figure S5

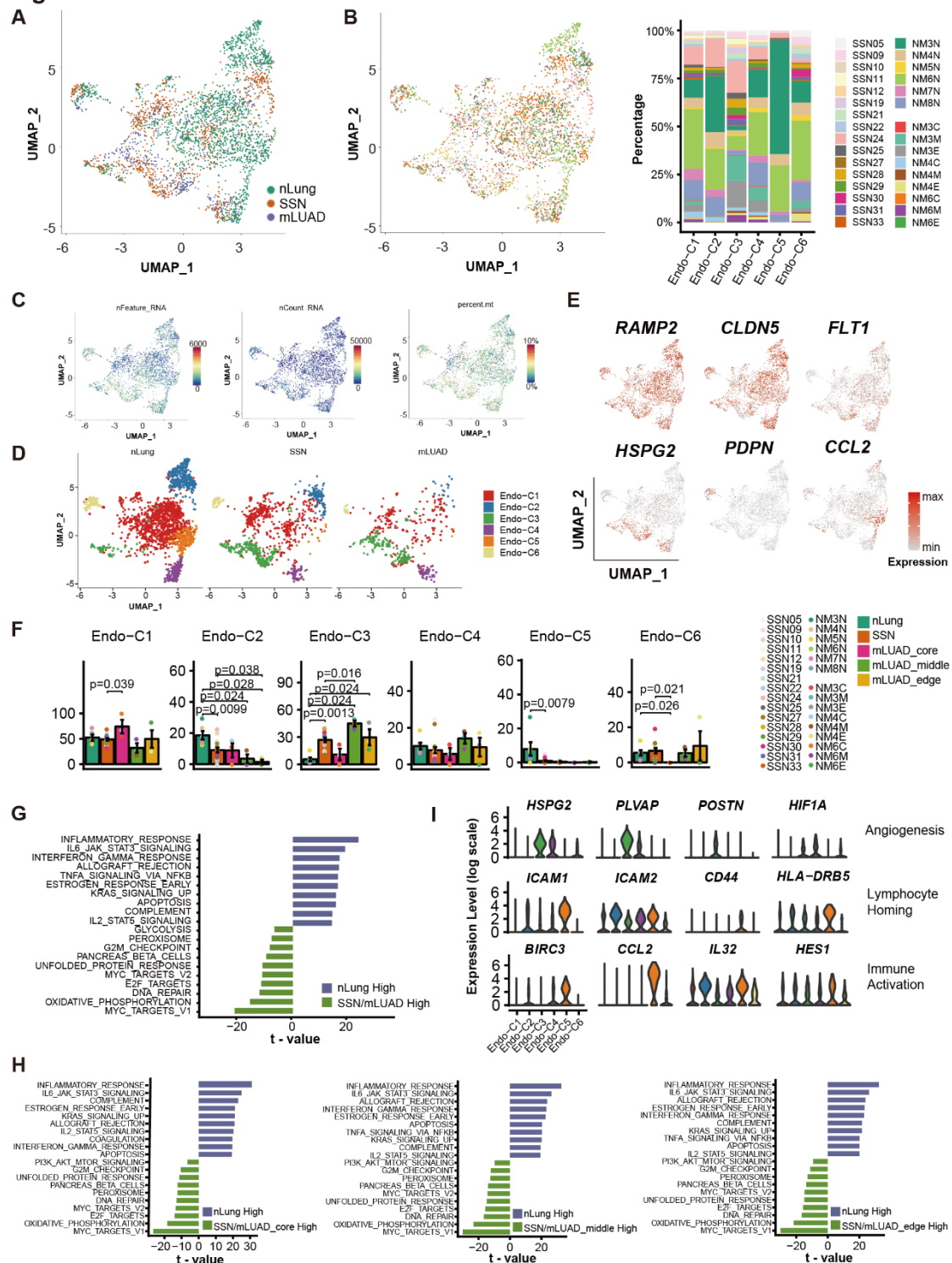


Figure S5. Endothelial cell subtypes in nLung, SSN, and mLUAD, Related to Figures 5A-5E.

A. Distribution of different groups in the UMAP plot. Each dot corresponds to a single cell, colored according to nLung, SSN and mLUAD, respectively. **B.** Sample distribution in the UMAP plot (left). Each bar corresponds to one cell type cluster, colored according to sample name (right). **C.** UMAP plot of cells colored by number of genes detected, UMIs RNA and percentage of mitochondrial genes. **D.** Annotation of the UMAP plots of endothelial cell

subtypes split by different groups. **E.** Canonical cell markers were used to identify endothelial cell subtypes. **F.** Percentages bar-plot similar to **Figure 5D**, but split mLUAD into different spatial positions. **G.** Differentially expressed pathways of endothelial cells in nLung compared with those in SSN/mLUAD as scored per cell by GSVA. The top 10 increased/decreased pathways are shown. **H.** Similar to (**G**), but split mLUAD into different spatial positions. **I.** Violin plots showing the expression of selected genes involved in angiogenesis, lymphocyte homing, and immune activation.

Figure S6

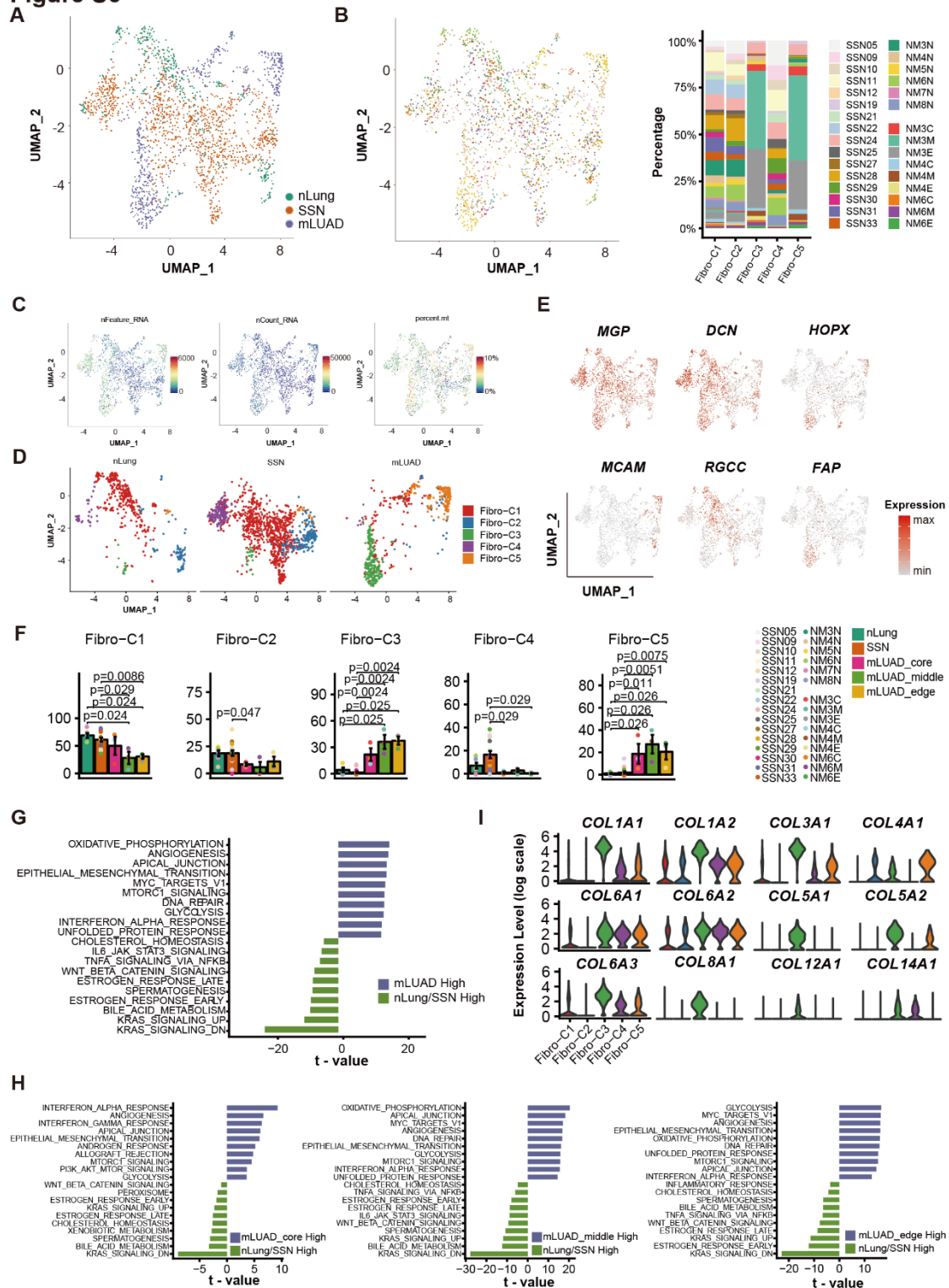


Figure S6. Fibroblast cell subtypes in nLung, SSN, and mLUAD, Related to Figures 5F-5K.

A. Distribution of different groups in the UMAP plot. Each dot corresponds to a single cell, colored according to nLung, SSN and mLUAD, respectively. **B.** Sample distribution in the UMAP plot (left). Each bar corresponds to one cell type cluster, colored according to sample name (right). **C.** UMAP plot of cells colored by number of genes detected, UMIs RNA and

percentage of mitochondrial genes. **D.** Annotation of the UMAP plots of fibroblasts subtypes split by different groups. **E.** Canonical cell markers were used to identify fibroblast subtypes. **F.** Percentages bar-plot similar to **Figure 5I**, but split mLUAD into different spatial positions. **G.** Differentially expressed pathways of fibroblasts in mLUAD compared with those in nLung/SSN as scored per cell by GSVA. The top 10 increased/decreased pathways are shown. **H.** Similar to (**G**), but split mLUAD into different spatial positions. **I.** Violin plots showing the expression of collagens in different fibroblast subtypes.

Figure S7

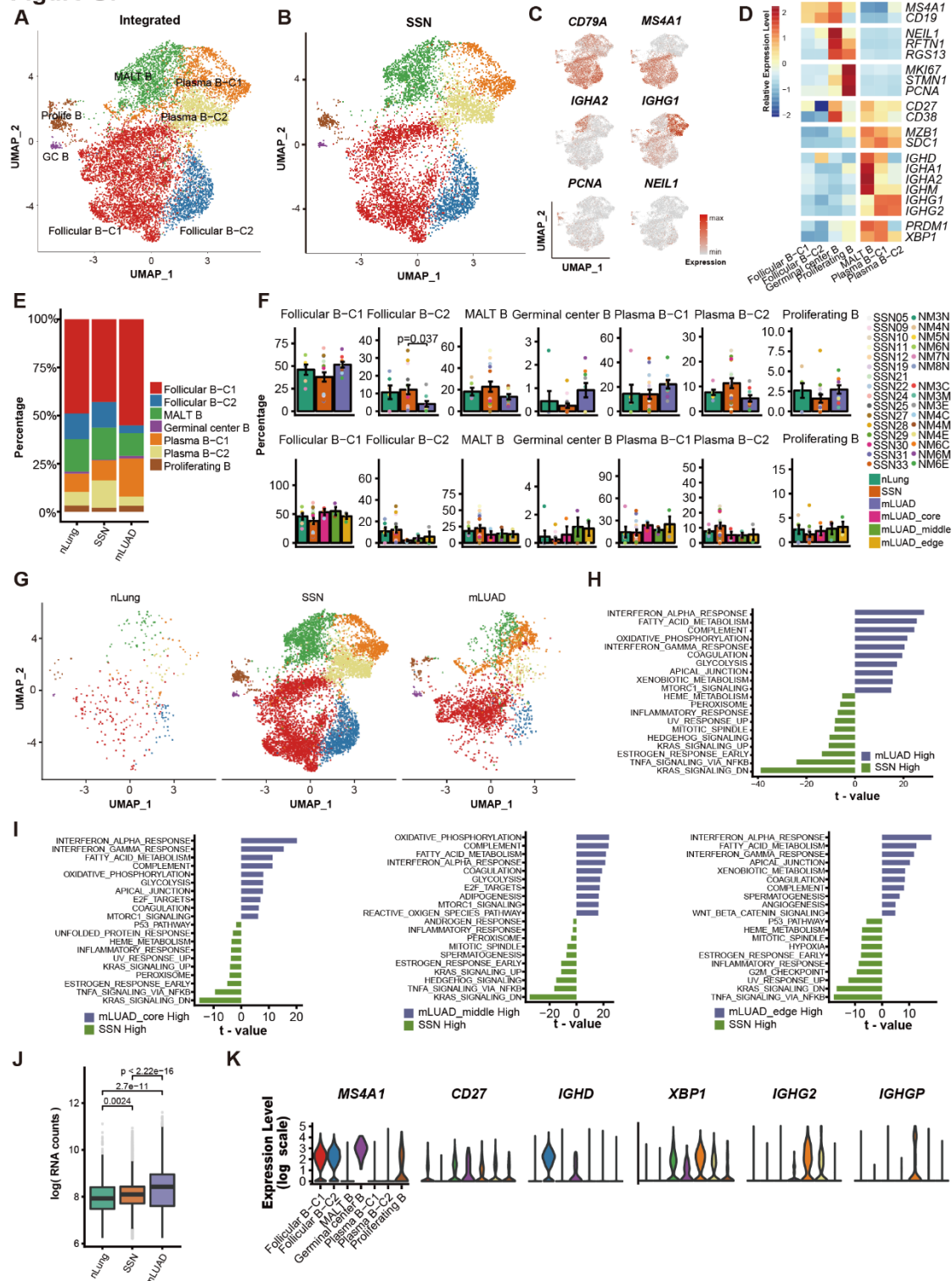


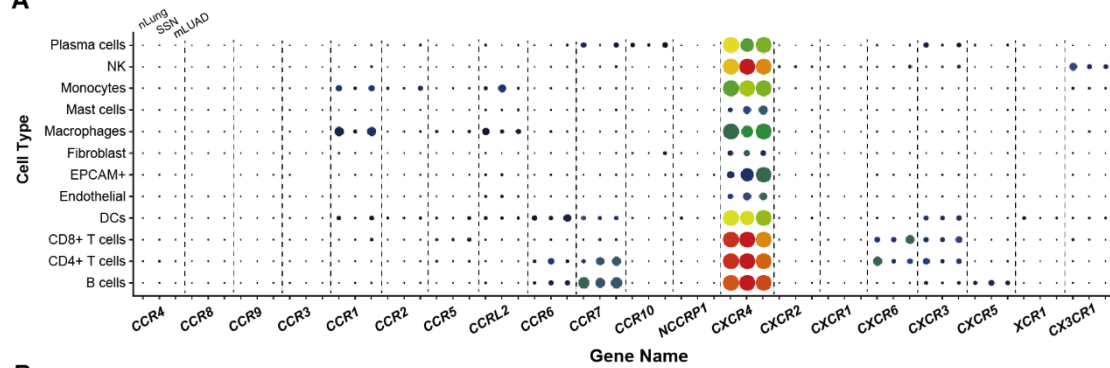
Figure S7. B cell subtypes in nLung, SSN, and mLUAD.

A. UMAP projection of 10,903 B cells, showing the composition of 7 subtypes. **B.** UMAP projection of 7,839 B cells derived from SSN. **C.** Canonical cell markers were used to identify B cell subtypes. **D.** Heatmap of marker gene expression in B-cell clusters. **E.** Average proportion of each B cell subtype between nLung, SSN, and mLUAD. **F.** Percentages of each B cell subtype in nLung, SSN, and mLUAD. The mLUAD also split into different spatial

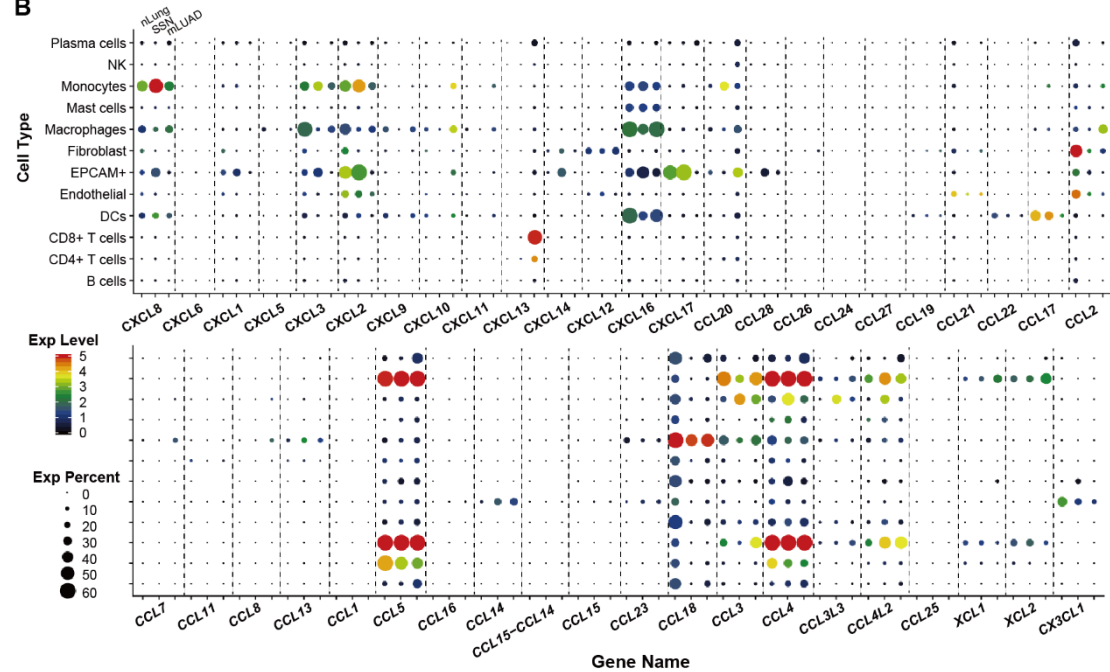
positions. Y axis: average percent of samples across the groups. Groups are shown in different colors. Each bar plot represents one subtype. Error bars represent $\pm$ s.e.m. Colored dots represent different samples. All differences with $P < 0.05$ are indicated; two-sided unpaired Wilcoxon rank-sum test was used for analysis. **G.** Annotation of the UMAP plots of B cell subtypes split by different groups. **H.** Differentially expressed pathways scored per cell by GSVA by comparing B cells in mLUAD to those in SSN. The top 10 increased/decreased pathways are shown. **I.** Similar to (**H**), but split mLUAD into different spatial positions. **J.** Boxplot showing the detected RNA abundance in nLung, SSN, and mLUAD. P-values were calculated by two-sided unpaired Wilcoxon rank-sum test. **K.** Violin plots showing the expression of selected marker genes of different B cell subtypes.

Figure S8

A



B



C

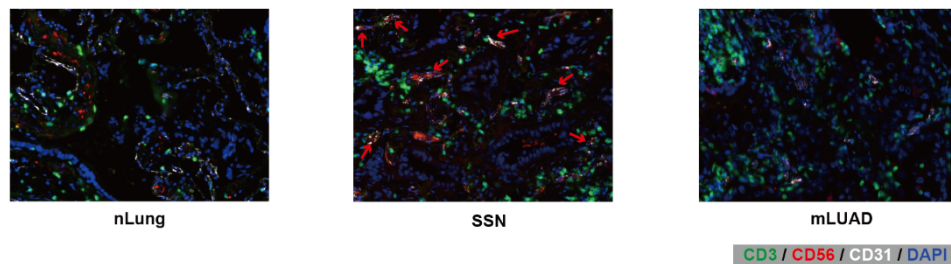


Figure S8. Expression of cellular ligands and receptors in different cell types, Related to Figure 6.

A and B. Dot plots showing the expression level and percentage of receptor genes (**A**) and ligand genes (**B**) in different cell types. The expression of each gene was split by the different groups. **C.** Immunostaining of endothelial cells and NK cells in nLung, SSN and mLUAD. Red arrows indicating the co-localized of NK and endothelial cells in SSN.

Supplementary Tables

Table S1. Characteristics of the 16 SSN patients included in this study.

Patient ID	Age	Sex	Location	Pathology	TNM	Stage	Smoking status	Recurrence	Survival
SSN05	55	Female	Right Upper	Invasive Adenocarcinoma	T1bN0M0	IA2	Never	No	Alive
SSN09	60	Female	Left Upper	Invasive Adenocarcinoma	T1bN0M0	IA2	Never	No	Alive
SSN10	85	Female	Left Upper	Carcinoma in situ	TisN0M0	0	Never	No	Alive
SSN11	46	Female	Left Upper	Invasive Adenocarcinoma	T1bN0M0	IA2	Never	No	Alive
SSN12	62	Female	Left Upper	Invasive Adenocarcinoma	T1bN0M0	IA2	Never	No	Alive
SSN19	52	Male	Left Upper	Invasive Adenocarcinoma	T1bN0M0	IA2	Never	No	Alive
SSN21	82	Male	Right Upper	Invasive Adenocarcinoma	T1cN0M0	IA3	Never	No	Alive
SSN22	53	Male	Right Lower	Invasive Adenocarcinoma	T1bN0M0	IA2	Never	No	Alive
SSN24	67	Male	Left Upper	Invasive Adenocarcinoma	T2aN0M0	IB	Active	No	Alive
SSN25	77	Male	Right Middle	Invasive Adenocarcinoma	T1cN0M0	IA3	Never	No	Alive
SSN27	66	Male	Left Upper	Invasive Adenocarcinoma	T1bN0M0	IA2	Never	No	Alive
SSN28	65	Female	Left Upper	Invasive Adenocarcinoma	T1cN0M0	IA3	Never	No	Alive
SSN29	74	Female	Left Upper	Invasive Adenocarcinoma	T1cN0M0	IA3	Never	No	Alive
SSN30	67	Female	Left Upper	Invasive Adenocarcinoma	T2aN0M0	IB	Never	No	Alive
SSN31	55	Female	Left Upper	Invasive Adenocarcinoma	T1bN0M0	IA2	Never	No	Alive
SSN33	73	Male	Right Upper	Invasive Adenocarcinoma	T1cN0M0	IA3	Former	No	Alive

Table S2. Characteristics of each subset for epithelial, immune and stromal cells.

Cluster Name	Annotated Name	Cells	nLung	SSN	mLUA D	Genes (mean)	UMIs (mean)	Mito (mean) %	Selected Marker Genes
AT1	Alveolar type I cell (AT1)	706	706	0	0	1286	3655	4.4	<i>AGER, CAV1, EMP2</i>
AT2	Alveolar type II cell (AT2)	1137	1137	0	0	1764	7211	3.8	<i>SFTPA1, SFTPA2, SFTPC, PGC</i>
Club	Secretory club cell	27	27	0	0	1894	9664	3.9	<i>TSPAN8, BPIFB1, SCGB1A1, TMEM45A</i>
Basal	Basal airway epithelial cells	13	13	0	0	441	684	5.1	<i>KRT6A, KRT17, KRT5</i>
Ciliated	Ciliated airway epithelial cells	114	114	0	0	2937	9548	5.7	<i>C1orf194, TMEM190, CAPS, TPPP3</i>
CD4-C1	Memory CD4 ⁺ T	7861	952	5754	1155	904	2906	3.1	<i>ANXA1, S100A4, CCR6, LTB, GPR183</i>
CD4-C2	Effector CD4 ⁺ T	5789	2089	3139	561	907	2477	3.2	<i>CD40LG, GZMA, GZMH, CXCR3, CXCR6, MHCII</i>
CD4-C3	Naïve CD4 ⁺ T	6497	423	5026	1048	935	3356	2.5	<i>SELL, CCR7, LEF1, TCF7</i>
CD4-C4	Suppressive Regulator T	3789	343	1852	1594	977	2794	2.9	<i>TNFRSF4, IL2RA, FOXP3, SELL, CTLA4, IKZF2</i>
CD4-C5	Exhausted CD4 ⁺ T	1471	35	684	752	1008	3022	3	<i>CXCL13, ICA1, TIGIT, TNFRSF4, BTLA</i>

CD8-C1	GZMK ⁺ Effector CD8 ⁺ T	9203	909	639 0	1904	799	2186	3.5	<i>GZMK, CCL4, CST7</i>
CD8-C2	ZNF683 ⁺ Effector CD8 ⁺ T	5762	740	358 5	1437	895	2378	3.6	<i>ZNF683, CCL5, HOPX, KLRC1</i>
CD8-C3	Exhausted CD8 ⁺ T	4188	39	314	3835	884	1915	2.5	<i>CXCL13, TIGIT, HAVCR2, LAG3, IFI6, IFNG</i>
CD8-C4	CX3CR1 ⁺ Effector CD8 ⁺ T	4169	807	281 9	543	904	2471	3.6	<i>NKG7, GNLY, FGFBP2, PRF1, CX3CR1, GZMH</i>
CD8-C5	Exhausted CD8 ⁺ T	1248	144	147	957	1576	4188	2.7	<i>STMN1, KIAA0101, TIGIT, CXCL13, IFNG</i>
NK-C1	CD3 ⁻ CD56 ^{dim} CD16 ⁺ NK	5627	990	431 9	318	964	2486	3.2	<i>FCGR3A, FGFBP2, CX3CR1, NKG7, TYROBP</i>
NK-C2	CD3 ⁻ CD56 ^{bright} CD16 ⁻ NK	1697	239	115 6	302	984	2846	3.4	<i>XCL1, XCL2, TYROBP, FCER1G, TRDC</i>
Macro-C1	Alveolar resident Macrophage	2737	1765	334	638	1798	6941	4.3	<i>MRC1, S100A6, EMP1, EMP3, SH3BGRL3, PPARG</i>
Macro-C2	Alveolar resident Macrophage	2195	1115	932	148	1490	7002	4	<i>SCD, RBP4, IL17RB, CAMP, GCHFR, PPARG, FABP4</i>
Macro-C3	Alveolar resident Macrophage	3364	2246	944	174	1983	9705	4.1	<i>CES1, FABP4, SLC19A3, PPARG, INHBA</i>
Macro-C4	Perivascular resident Macrophage	1394	273	953	168	965	3410	3.3	<i>LYVE1, LILRB5, SEPP1, FOLR2, SLC40A1</i>
Macro-C5	Anti-inflammatory macrophages	1803	1013	374	416	2147	9893	4.2	<i>MMP7, TIMP3, PLA2G7, CHI3L1, CTSB</i>
Macro-C6	Anti-inflammatory macrophages	906	197	482	227	1739	6676	3.4	<i>TNF, ADAMDEC1, HS3ST1, RGS1, C3, AXL</i>

Macro-C7	Tumor-associated Macrophage	324	8	1	315	1113	3574	3.6	<i>SPPI, SLAMF9, LDHA, VEGFA, SLC2A3</i>
Macro-C8	Proliferating Macrophage	711	149	449	113	1682	7361	3.4	<i>STMN1, H2AFZ, TUBA1B, PCNA</i>
Macro-C9	Early-stage Macrophage	760	429	139	192	1707	7134	4.1	<i>CXCL10, GBP1, CXCL9, CXCL11</i>
Mono-C1	Classical CD14 ⁺ Monocyte	1308	638	377	293	1095	3530	4.3	<i>CD14, FCN1, S100A12, VCAN, SERPINB2</i>
Mono-C2	Non-classical CD16 ⁺ Monocyte	496	281	148	67	1196	3705	4.5	<i>FCGR3A, CDKN1C, POU2F2, ZNF703</i>
Mono-DCs	Monocytes Derived DC	1068	296	594	178	1780	7450	3.2	<i>CD14, FCGR2B, CCL17, CLEC10A</i>
DCs-C1	Conventional Dendritic cell type 1	378	87	260	31	1020	4421	2	<i>CLEC9A, BATF3, IRF8, CPVL, CADM1</i>
DCs-C2	Conventional Dendritic cell type 2	361	24	272	65	849	3031	2.9	<i>CD1C, FCER1A, FCGBP, CD1A, CD207, HLA-DQB2</i>
DCs-C3	Migratory Conventional Dendritic cell	236	48	167	21	870	3551	2.8	<i>FSCN1, CCR7, LAMP3, CCL19, CCL22, CD40, BIRC3</i>
pDCs	Plasmacytoid Dendritic cell	247	52	152	43	1646	6033	2.3	<i>IRF4, LILRA4, TCF4, MZB1</i>
Granulocytes	Granulocyte	92	8	77	7	363	786	3	<i>G0S2, FCGR3B, S100A8, S100A9</i>
Endo-C1	Extra-alveolar Capillary Endothelial cell	1696	1066	432	198	764	1691	2.9	<i>EDN1, SLC6A4, FCN3, CD36, CA4, SLC6A4</i>
Endo-C2	Alveolar Capillary Endothelial cell	528	385	125	18	958	2227	2.7	<i>HPGD, EDNRB, IL1RL1, ICAM2</i>
Endo-C3	Tumor Endothelial cell	497	75	247	175	1712	5263	3.8	<i>HSPG2, IGFBP7, PLVAP, SPRY1, VWA1, ESM1, NID2, ACKR1, SELP</i>
Endo-C4	Arterial Endothelial cell	273	165	56	52	1309	3496	3.3	<i>GJA5, FBLN5, DKK2, IGFBP3</i>

Endo-C5	Extra-alveolar Capillary Normal Endothelial cell	238	224	12	2	860	2013	2.6	<i>EDN1, CCL2, AKAP12, CSF3, IL6, ICAM1</i>
Endo-C6	Lymphatic Endothelial cell	149	93	39	17	1762	5457	3.6	<i>CCL21, TFF3, NRP2, PDPN</i>
Fibro-C1	Normal Fibroblast	1216	311	773	132	751	1991	3.3	<i>RGCC, MACF1, A2M, DST, NPNT</i>
Fibro-C2	Myofibroblast	334	92	211	31	846	2737	3.1	<i>ACTA2, MYL9, PTN, MT2A, FAM162B, KCNK3</i>
Fibro-C3	Cancer-associated Fibroblast	286	14	22	250	1870	5900	4.3	<i>COL3A1, COL1A1, COL1A2, POSTN, FAP, TGFB1</i>
Fibro-C4	Immune-modulatory Fibroblast	214	39	170	5	1777	5339	2.9	<i>OGN, GSN, CFD, CXCL14</i>
Fibro-C5	Pericyte	207	8	20	179	1471	4169	4.3	<i>RGS5, NOTCH3, PDGFRB, PTP4A3, COL4A2, MMP9, DES</i>
Follicular B-C1	Memory B cell	5034	150	336 6	1518	1014	3781	3.3	<i>MS4A1, CD27, LTB, GPR183</i>
Follicular B-C2	Naïve B cell	1193	41	104 1	111	929	3148	3.4	<i>MS4A1, IGHD, FCER2, CXCR4</i>
MALT B	MALT B cell	1686	52	130 7	327	774	6718	1.4	<i>JCHAIN, IGHA1, IGHA2, IGHM</i>
Germinal center B	Germinal center B cell	66	3	28	35	1636	5214	2.6	<i>NEIL1, TCL1A, RGS13, RFTN1</i>
Plasma B-C1	Plasma IgG mature cell	1393	29	816	548	1029	11746	1.3	<i>MZB1, IGHG1, IGHG2, IGHG3, XBPI, PRDMI</i>
Plasma B-C2	Plasma IgG immature cell	1265	22	111 2	131	483	4010	2.5	<i>MZB1, IGHG1, IGHG2, IGHG4</i>

Proliferating B	Proliferating B cell	266	10	169	87	1381	6487	2.1	<i>STMN1, MKI67, PCNA</i>
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Table S3. Differential cell population abundance by differential abundance (DA) analyses.

SSN vs nLung	Cluster Name	logF C	logCP M	F	Pvalu e	FDR	SSN vs mLUAD	Cluster Name	logFC	logCP M	F	Pvalu e	FDR
Overall	T cells	0.584	18.855	4.894	3.70E -02	4.44E -02	Overall	T cells	-0.285	19.101	2.713	1.12E -01	2.23E -01
	NK	0.106	17.136	0.028	8.69E -01	8.69E -01		NK	1.527	16.781	7.282	1.22E -02	3.65E -02
	Myeloid	- 1.729	17.870	14.65 4	8.48E -04	1.70E -03		Myeloid	-0.119	17.210	0.067	7.98E -01	7.98E -01
	Mast cells	2.168	15.156	20.40 0	1.51E -04	5.43E -04		Mast cells	1.541	15.116	11.256	2.48E -03	1.49E -02
	B cells	2.428	15.989	7.180	1.33E -02	2.00E -02		B cells	-0.173	16.412	0.077	7.84E -01	7.98E -01
	Plasma cells	3.746	15.399	19.76 2	1.81E -04	5.43E -04		Plasma cells	0.329	15.705	0.345	5.62E -01	7.98E -01
T/NK	CD4-C1	0.519	17.230	5.832	2.36E -02	4.71E -02	T/NK	CD4-C1	0.945	17.082	18.697	1.80E -04	4.31E -04
	CD4-C2	- 1.360	17.013	34.94 8	3.95E -06	2.37E -05		CD4-C2	1.407	16.143	39.583	8.90E -07	3.56E -06
	CD4-C3	1.431	16.897	28.62 3	1.62E -05	6.50E -05		CD4-C3	0.946	16.893	11.052	2.53E -03	3.80E -03
	CD4-C4	0.505	15.702	2.191	1.52E -01	2.06E -01		CD4-C4	-1.046	16.282	16.702	3.46E -04	6.92E -04

	CD4-C5	2.234	14.020	8.982	6.25E-03	1.50E-02		CD4-C5	-1.424	14.986	12.112	1.70E-03	2.92E-03
	CD8-C1	0.586	17.371	10.263	3.75E-03	1.13E-02		CD8-C1	0.441	17.365	8.883	5.95E-03	7.93E-03
	CD8-C2	0.055	16.615	0.037	8.49E-01	8.51E-01		CD8-C2	-0.189	16.700	0.491	4.89E-01	4.89E-01
	CD8-C3	0.852	13.084	4.411	4.62E-02	7.92E-02		CD8-C3	-4.694	16.416	177.811	1.81E-13	2.17E-12
	CD8-C4	-0.696	16.383	2.159	1.55E-01	2.06E-01		CD8-C4	0.757	15.939	3.517	7.15E-02	8.58E-02
	CD8-C5	-2.110	13.051	40.128	1.39E-06	1.67E-05		CD8-C5	-3.792	14.549	114.760	2.81E-11	1.68E-10
	NK-C1	-0.423	16.946	1.019	3.23E-01	3.87E-01		NK-C1	2.091	16.362	22.578	5.80E-05	1.74E-04
	NK-C2	-0.057	15.042	0.036	8.51E-01	8.51E-01		NK-C2	0.262	14.942	0.574	4.55E-01	4.89E-01
Myeloid	Macro-C1	-1.857	16.446	56.060	3.65E-08	3.10E-07	Myeloid	Macro-C1	-1.937	16.643	59.394	1.09E-09	9.24E-09
	Macro-C2	0.091	16.967	0.106	7.48E-01	7.48E-01		Macro-C2	1.542	16.647	26.096	6.78E-06	3.01E-05
	Macro-C3	-0.974	17.380	13.834	8.84E-04	1.37E-03		Macro-C3	1.080	16.782	11.799	1.31E-03	2.78E-03
	Macro-C4	1.755	16.626	22.719	5.21E-05	1.48E-04		Macro-C4	1.327	16.643	15.515	2.89E-04	7.03E-04
	Macro-C5	-1.053	16.161	9.207	5.15E-03	7.29E-03		Macro-C5	-1.251	16.353	16.622	1.89E-04	5.35E-04

	Macro-C6	1.685	15.855	15.61 0	4.77E -04	9.01E -04		Macro-C6	-0.047	16.241	0.026	8.72E -01	8.72E -01
	Macro-C7	- 1.613	11.659	5.981	3.33E -02	4.04E -02		Macro-C7	-7.688	15.231	226.11 8	6.35E -19	1.08E -17
	Macro-C8	1.682	15.738	35.48 4	2.03E -06	1.15E -05		Macro-C8	0.799	15.891	6.844	1.22E -02	1.88E -02
	Macro-C9	- 1.201	14.887	16.96 5	3.04E -04	6.46E -04		Macro-C9	-1.612	15.222	25.959	7.08E -06	3.01E -05
	Mono-C1	- 0.674	16.361	2.708	1.11E -01	1.26E -01		Mono-C1	-0.644	16.416	4.269	4.47E -02	5.85E -02
	Mono-C2	- 0.214	14.909	0.251	6.20E -01	6.59E -01		Mono-C2	0.112	14.911	0.073	7.89E -01	8.38E -01
	Mono-DCs	1.565	16.401	28.75 3	1.02E -05	4.34E -05		Mono-DCs	0.794	16.512	7.023	1.11E -02	1.88E -02
	DCs-C1	2.015	15.151	25.29 3	2.54E -05	8.63E -05		DCs-C1	2.068	15.193	22.080	2.60E -05	8.85E -05
	DCs-C2	3.618	14.900	62.75 6	1.23E -08	2.08E -07		DCs-C2	0.775	15.255	3.927	5.38E -02	6.53E -02
	DCs-C3	2.056	14.295	17.18 3	2.83E -04	6.46E -04		DCs-C3	1.606	14.495	11.394	1.55E -03	2.93E -03
	pDCs	1.908	14.614	8.834	6.20E -03	8.11E -03		pDCs	1.009	14.839	2.985	9.48E -02	1.07E -01
Endothelia 1	Granulocyte s	3.247	13.386	14.32 4	7.93E -04	1.35E -03	Endothelia 1	Granulocyte s	2.058	13.780	4.969	3.40E -02	4.82E -02
	Endo-C1	- 0.123	18.930	0.441	5.13E -01	6.15E -01		Endo-C1	0.050	18.880	0.050	8.24E -01	8.24E -01

	Endo-C2	- 0.915	17.140	7.433	1.13E -02	2.26E -02		Endo-C2	1.168	16.744	4.171	4.97E -02	2.05E -01
	Endo-C3	2.720	17.647	63.06 7	1.98E -08	1.19E -07		Endo-C3	-0.235	18.232	0.706	4.07E -01	6.11E -01
	Endo-C4	- 0.426	16.452	1.541	2.25E -01	3.38E -01		Endo-C4	-0.688	16.753	3.500	7.08E -02	2.05E -01
	Endo-C5	- 2.807	15.924	23.09 6	5.57E -05	1.67E -04		Endo-C5	1.148	15.365	2.826	1.03E -01	2.05E -01
	Endo-C6	0.242	15.996	0.222	6.42E -01	6.42E -01		Endo-C6	0.408	16.357	0.383	5.40E -01	6.48E -01
Fibroblast	Fibro-C1	- 0.163	19.259	0.785	5.89E -01	9.42E -01	Fibroblast	Fibro-C1	0.809	18.998	6.732	1.45E -02	1.45E -02
	Fibro-C2	- 0.031	17.614	0.005	9.42E -01	9.42E -01		Fibro-C2	1.297	17.352	6.780	1.42E -02	1.45E -02
	Fibro-C3	- 1.172	15.513	1.247	2.76E -01	6.90E -01		Fibro-C3	-4.153	17.082	46.576	1.39E -07	6.94E -07
	Fibro-C4	1.092	17.269	2.593	1.22E -01	6.08E -01		Fibro-C4	3.782	17.053	21.627	6.17E -05	1.03E -04
	Fibro-C5	0.110	15.305	0.010	9.20E -01	9.42E -01		Fibro-C5	-3.692	16.752	34.917	1.76E -06	4.40E -06
B/Plasma	Follicular B-C1	- 0.292	18.626	0.387	5.38E -01	9.11E -01	B/Plasma	Follicular B-C1	-0.432	18.716	1.344	2.54E -01	2.97E -01
	Follicular B-C2	0.056	16.921	0.006	9.38E -01	9.38E -01		Follicular B-C2	1.549	16.542	5.874	2.07E -02	6.96E -02
	MALT B	0.289	17.737	0.209	6.51E -01	9.11E -01		MALT B	0.750	17.571	2.080	1.58E -01	2.77E -01

	Germinal center B	- 1.275	13.129	1.379	2.49E-01	8.81E-01		Germinal center B	-1.527	13.298	5.133	2.98E-02	6.96E-02
	Plasma B-C1	0.210	17.025	0.052	8.21E-01	9.38E-01		Plasma B-C1	-0.735	17.366	1.383	2.48E-01	2.97E-01
	Plasma B-C2	0.704	16.776	1.239	2.74E-01	8.81E-01		Plasma B-C2	1.199	16.582	5.362	2.66E-02	6.96E-02
	Proliferating B	- 0.734	14.652	0.800	3.78E-01	8.81E-01		Proliferating B	-0.572	14.691	1.029	3.17E-01	3.17E-01